

Short Questions

Question 1:

List out controls which does not have events.

List out controls which does not have events.

- ✓ Controls in VB are mostly associated with different kind of events. But there are some controls which does not have any events. Here is list of those controls.
 - ☐ Label controls
 - ☐ Image controls
 - ☐ Timer controls
 - ☐ Div controls
 - ☐ Line controls
 - ☐ shape controls

Question 2:

What is early and late binding?

What is early and late binding?

Early Binding:

- ✓ Early binding refers to the assignment of values to variables during design time

Late Binding:

- ✓ Late Binding refers to the assignment of values to variables during run time.

Question 3:

How can we repeat same task 10 times?

How can we repeat same task 10 times?

- ✓ Loops are the type of control structures that repeat a statement or set of statements for a number of times or according to condition satisfied.
- ✓ So in case we want to repeat same task 10 times, FOR Loop is best option over there.

Question 4:

Differentiate between do-while and for loop.

Differentiate between do-while and for loop.

For Loop	Do While Loop
Statement(s) is executed once the condition is checked.	Condition is checked after the statement(s) is executed.
It might be that statement(s) gets executed zero times.	Statement(s) is executed at least once.
Initialization may be outside or in condition box.	Initialization may be outside or within the loop.
for loop is entry-controlled loop.	do-while is exit controlled loop.

Question 5:

What is difference between ListBox and ComboBox?

What is difference between ListBox and ComboBox?

- ✓ The ComboBox and the ListBox controls have similar behaviors, and in some cases may be interchangeable. There are times, however, when one or the other is more appropriate to a task.
- ✓ Generally, a combo box is appropriate when there is a list of suggested choices, and a list box is appropriate when you want to limit input to what is on the list. A combo box contains a text box field, so choices not on the list can be typed in.

Question 6:

Describe the benefits of array with help of code.

Describe the benefits of array with help of code.

Some advantages of arrays are as follows:

- ✓ Arrays can store large number of values with single name.
- ✓ Arrays are used to process many values easily and quickly.
- ✓ The Values stored in an array can be sorted easily.
- ✓ A search process can be applied on arrays easily.

Private Sub Command1_Click()

Dim Marks(4) As Integer

For i = 0 To 4

 Marks(i) = InputBox("Enter marks: ")

Next

For i = 0 To 4

 Print "Value at index " & i & " is " & Marks(i)

Next

End Sub

Question 7:

Define a function which takes two variables (named as x and y) as integer and return a solution of the following equation. (Only function code is required here). $X^2 + Y^2$

Define a function which takes two variables (named as x and y) as integer and return a solution of the following equation. (Only function code is required here). $X^2 + Y^2$

```
Function Square(x As Integer, y As Integer)
```

```
    Dim n As Integer
```

```
    n = x*x + y*y
```

```
    Square = n
```

```
End Function
```

Question 8:

What is the substitute of Select Case Statement? Give an example code.

What is the substitute of Select Case Statement? Give an example code.

- ✓ Nested-IF structure can be used as substitute of select case statements.

```
choice = 3
```

```
IF choice = 1 THEN
```

```
    Print "Good Morning"
```

```
ELSE
```

```
    IF choice = 2 THEN
```

```
        Print "Good Afternoon"
```

```
    ELSE
```

```
        IF choice = 3 THEN
```

```
            Print "Good Evening"
```

```
        ELSE
```

```
            Print "Invalid Choice"
```

```
        END IF
```

```
    END IF
```

```
END IF
```

Question 9:

What are different data types in programming?

What are different data types in programming?

✓ The following table shows the Visual Basic data types:

- ☐ Boolean
- ☐ Byte
- ☐ Char
- ☐ Date
- ☐ Decimal
- ☐ Double
- ☐ Integer
- ☐ Short
- ☐ Single
- ☐ String

Question 10:

Define the type of errors and how can we remove them?

Define the type of errors and how can we remove them?

- ✓ Error may occur during execution even if the program is designed very carefully.

Types of errors:

- ☐ Syntax errors
 - ☐ Logical errors
 - ☐ Run-time errors
- ✓ Visual basic provide facility to handle such errors. Visual basic include a default **Error Handler** that handles the situation when an error occurs. The error handler performs the following steps in case of an error
 - ☐ It displays the error number and short description of the error
 - ☐ It terminates the application.

Long Question

Question 1:

What are relational and logical operators? Describe with example code.

Relational Operators in VB

- ✓ The relational operators are used to specify conditions in program.
- ✓ A relational operator compares two values.
- ✓ It produce results as true or false.
- ✓ The relational operators are sometimes called conditional operators or comparison operators as they test conditions that are either true or false.
- ✓ Visual Basic provides the following relational operators:
- ✓ `>` , `<` , `>=` , `<=` , `==` , `<>`

Relational Operators in VB

Operator	Description
>	Greater than operator returns true if the value on the left side of > is greater than the value on the right side. Otherwise returns false .
<	Less than operator return true if the value of the left side of < is less than the value on the right side. Otherwise return false .
==	Equal to operator return true if the value of the left side of == is equal to the value on the right side. Otherwise return false .
>=	Greater than or equal to operator return true if the value of the left side of >= is greater than or equal to the value on the right side. Otherwise return false .
<=	Less than or equal to operator return true if the value of the left side of >= is less than or equal to the value on the right side. Otherwise return false .
<>	The not equal to operator return true if the value of the left side of <> is not equal to the value on the right side. Otherwise return false .

Logical Operators

- ✓ Logical operators are used to evaluate compound conditions. There are three logical operators.
 - ✓ AND operator
 - ✓ OR operator
 - ✓ NOT operator

Logical Operators

- ✓ **AND Operator:**
- ✓ The AND operator is used to evaluate two conditions.
- ✓ It produce **true** result if both conditions are true. It produce **false** result if any one condition is false.

Condition 1	Operator	Condition 2	Result
False	AND	False	False
False	AND	True	False
True	AND	False	False
True	AND	True	True

Logical Operators

OR Operator:

- ✓ The OR operator is used to evaluate two conditions.
- ✓ It produce true result if either condition is true. It produce false result if both conditions are false.

Condition 1	Operator	Condition 2	Result
False	OR	False	False
False	OR	True	True
True	OR	False	True
True	OR	True	True

Logical Operators

NOT Operator:

- ✓ The NOT operator is used to reverse the result of a condition. It produces true result if the condition is false. It produces false result if condition is true.

Operator	Condition 2	Result
Not	False	True
Not	True	False

Question 2:

Write algorithm to print table from 2 to 10.

Program Solution

```
Private Sub Command1_click ()
```

```
For a = 2 To 10
```

```
    For b = 1 to 10
```

```
        Print a & "*" & b & "=" & a*b
```

```
    Next
```

```
Next
```

```
End Sub
```

Question 3:

Differentiate between argument pass by value and pass by reference.

PASS BY VALUE

Mechanism of copying the function parameter value to another variable

Changes made inside the function are not reflected in the original value

Makes a copy of the actual parameter

Function gets a copy of the actual content

Requires more memory

Requires more time as it involves copying values

PASS BY REFERENCE

Mechanism of passing the actual parameters to the function

Changes made inside the function are reflected in the original value

Address of the actual parameter passes to the function

Function accesses the original variable's content

Requires less memory

Requires a less amount of time as there is no copying

Question 4:

Write a program which takes 6-digit number and display each digit on separate line.

Program

```
Private Sub Command1_Click()
```

```
Dim num As Integer
```

```
num = val(InputBox("Enter a 6-digit number: "))
```

```
While num >= 0
```

```
    Print num Mod 10
```

```
    num = num/10
```

```
Wend
```

```
End Sub
```

Question 5:

Write a program which input numbers and display average of two floating point.

Program

```
Private Sub Command1_Click()
```

```
Dim n1 As Single, n2 As Single, avg As Single
```

```
n1 = val(InputBox("Enter first floating point number: "))
```

```
n2 = val(InputBox("Enter first floating point number: "))
```

```
avg = (n1 + n2)/2
```

```
Print "Average = " & avg
```

```
End Sub
```